

Transportation Model Example 1

Problem Definition and Data (Solved)

How many tons of wheat to transport from each grain elevator to each mill on a monthly basis in order to minimize the total cost of transportation?

<u>Grain Elevator</u>	<u>Supply</u>	<u>Mill</u>	<u>Demand</u>
1. Kansas City	150	A. Chicago	220
2. Omaha	175	B. St. Louis	100
3. Des Moines	275	C. Cincinnati	300
Total	600 tons	Total	600 tons

<u>Transport Cost from Grain Elevator to Mill (\$/ton)</u>			
<u>Grain Elevator</u>	<u>A. Chicago</u>	<u>B. St. Louis</u>	<u>C. Cincinnati</u>
1. Kansas City	\$ 6	\$ 8	\$ 10
2. Omaha	7	11	11
3. Des Moines	4	5	12

Transportation Model Example

Transportation Network Routes

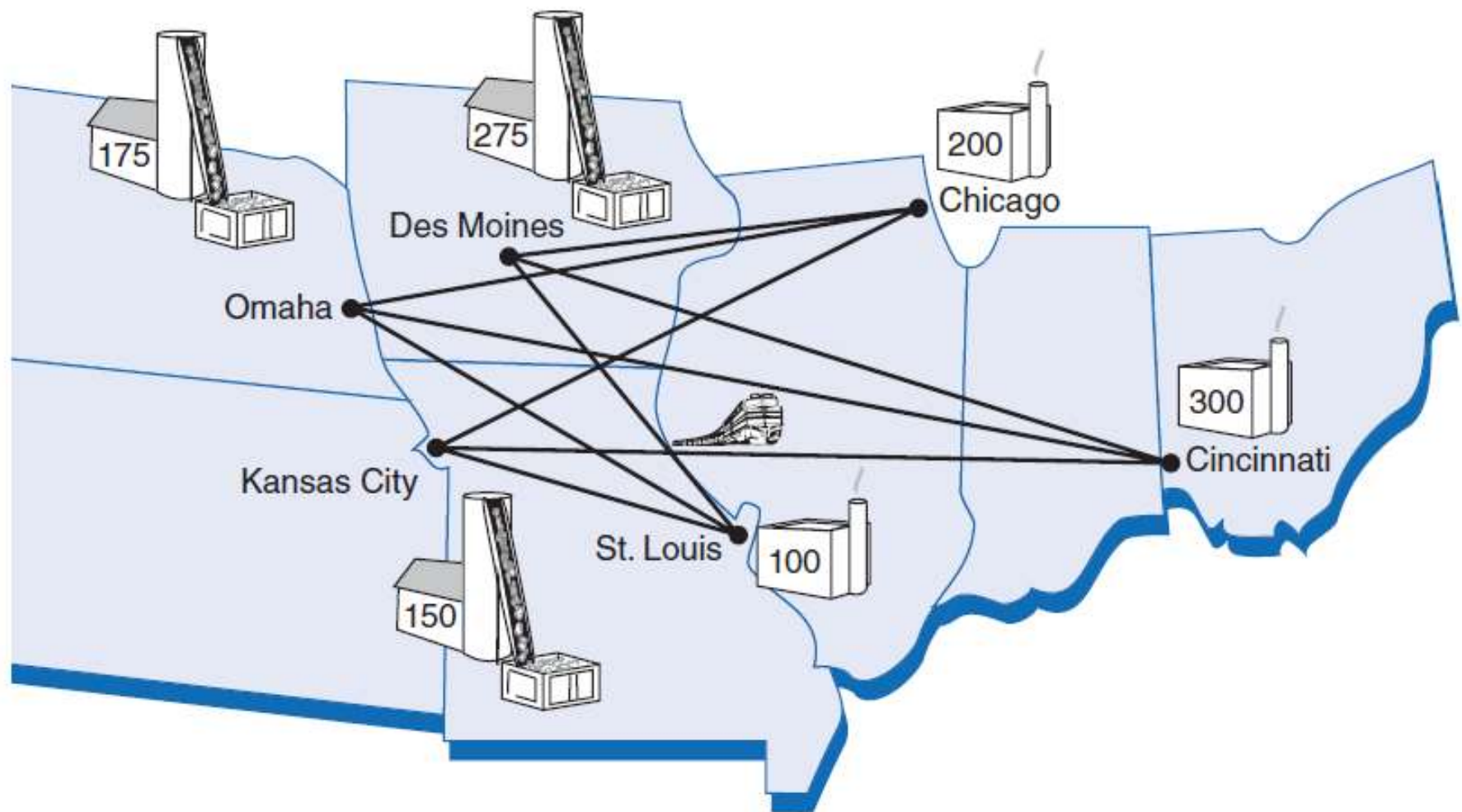


Figure 6.1 Network of transportation routes for wheat shipments

Transportation Model Example

Model Formulation

$$\text{Minimize } Z = \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C}$$

subject to:

$$x_{1A} + x_{1B} + x_{1C} = 150$$

$$x_{2A} + x_{2B} + x_{2C} = 175$$

$$x_{3A} + x_{3B} + x_{3C} = 275$$

$$x_{1A} + x_{2A} + x_{3A} = 200$$

$$x_{1B} + x_{2B} + x_{3B} = 100$$

$$x_{1C} + x_{2C} + x_{3C} = 300$$

$$x_{ij} \geq 0$$

x_{ij} = tons of wheat from each grain elevator, i , $i = 1, 2, 3$,
to each mill j , $j = A, B, C$

Transportation Model Example

Computer Solution with Excel (1 of 4)

The screenshot shows an Excel spreadsheet titled "Exhibit6.1 - Microsoft Excel". The formula bar at the top displays $=\text{SUMPRODUCT}(C5:E7,K5:M7)$, which is identified by a callout as the "Objective function".

The spreadsheet contains two tables. The first table, "The Wheat Shipping Example", is located in cells A3:M7. The second table, "Shipping costs (\$/ton)", is located in cells K3:M7. Both tables have columns for "Grain Elevators" (Chicago, St. Louis, Cincinnati) and "Mills".

Callouts identify key components of the model:

- Objective function:** $=\text{SUMPRODUCT}(C5:E7,K5:M7)$
- Decision variables in cells C5:E7:** The values in cells C5, D5, and E5 (150, 175, 275) are identified as decision variables.
- Cost array in cells K5:M7:** The values in cells K5, L5, and M5 (6, 8, 10) are identified as the cost array.
- Formula:** $=C7+D7+E7$ is shown in a callout, pointing to the formula in cell C7.
- Formula:** $=D5+D6+D7$ is shown in a callout, pointing to the formula in cell D5.

Grain Elevators	Chicago	St. Louis	Cincinnati	Supply	Grain Shipped
Kansas City	25	0	125	150	150
Omaha	0	0	175	175	175
Des Moines	175	100	0	275	275
Demand	200	100	300	600	
Grain Shipped	200	100	300		
Cost =	4525				

Grain Elevators	Chicago	St. Louis	Cincinnati
Kansas City	6	8	10
Omaha	7	11	11
Des Moines	4	5	12

Exhibit 6.1

Transportation Model Example

Computer Solution with Excel (2 of 4)

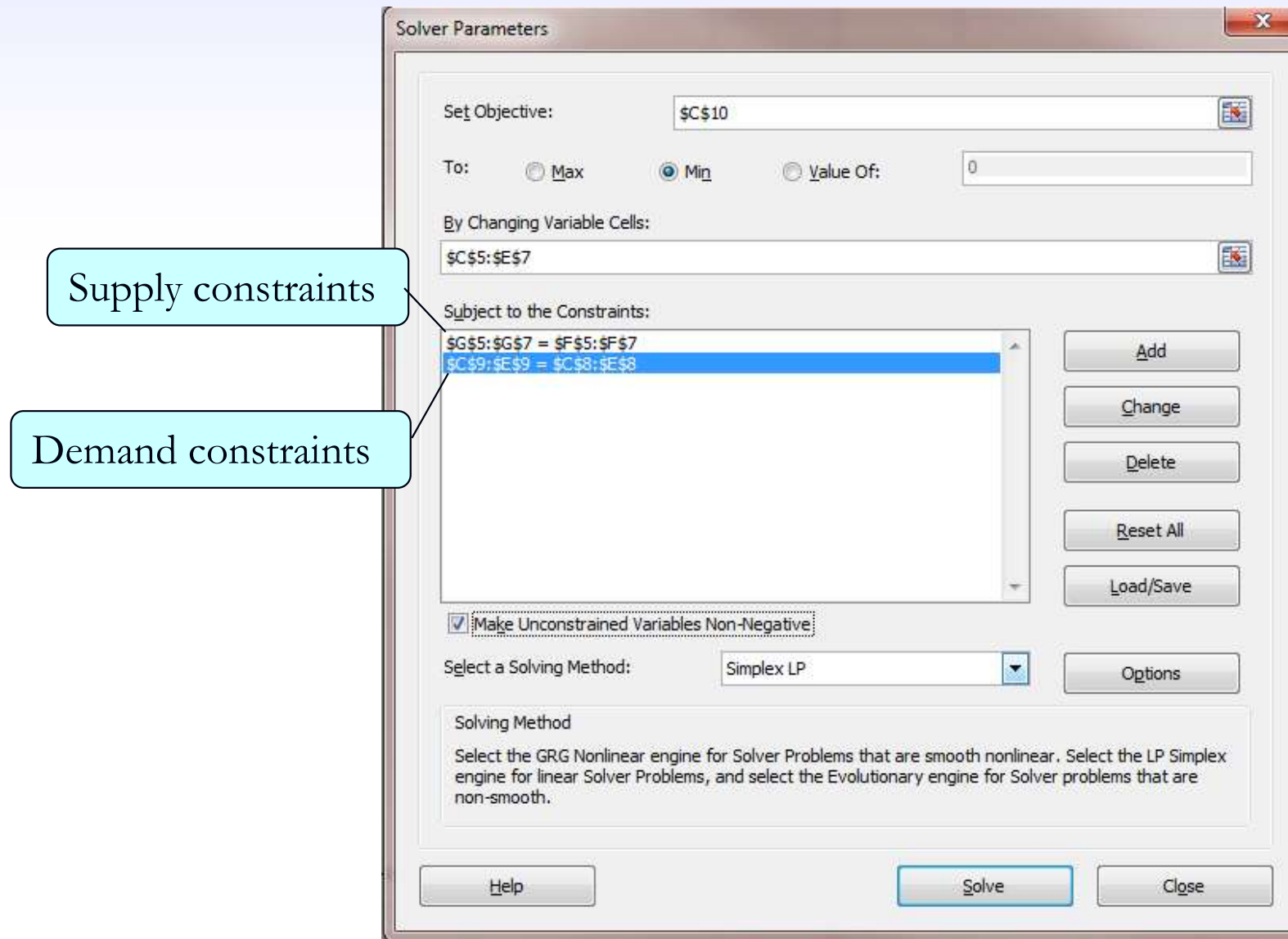


Exhibit 6.2

Transportation Model Example

Computer Solution with Excel (3 of 4)

Exhibit6.1 [Compatibility Mode] - Microsoft Excel

File Home Insert Page Layout Formulas Data Review View

From Access From Web From Text From Other Sources Existing Connections Refresh All Connections Sort Filter Clear Reapply Advanced Text to Columns Remove Duplicates Data Validation Consolidate What-If Analysis Group Ungroup Subtotal Outline

C10 fx =SUMPRODUCT(C5:E7,K5:M7)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	The Wheat Shipping Example													
2									Shipping costs (\$/ton)					
3				<i>Mills</i>			<i>Grain</i>					<i>Mills</i>		
4	<i>Grain Elevators</i>	Chicago	St. Louis	Cincinnati	<i>Supply</i>	<i>Shipped</i>			<i>Grain Elevators</i>	Chicago	St. Louis	Cincinnati		
5	Kansas City	25	0	125	150	150			Kansas City	6	8	10		
6	Omaha	0	0	175	175	175			Omaha	7	11	11		
7	Des Moines	175	100	0	275	275			Des Moines	4	5	12		
8	<i>Demand</i>	200	100	300	600									
9	<i>Grain Shipped</i>	200	100	300										
10	<i>Cost =</i>	4525												
11														

Exhibit 6.3

Transportation Model Example

Computer Solution with Excel (4 of 4)

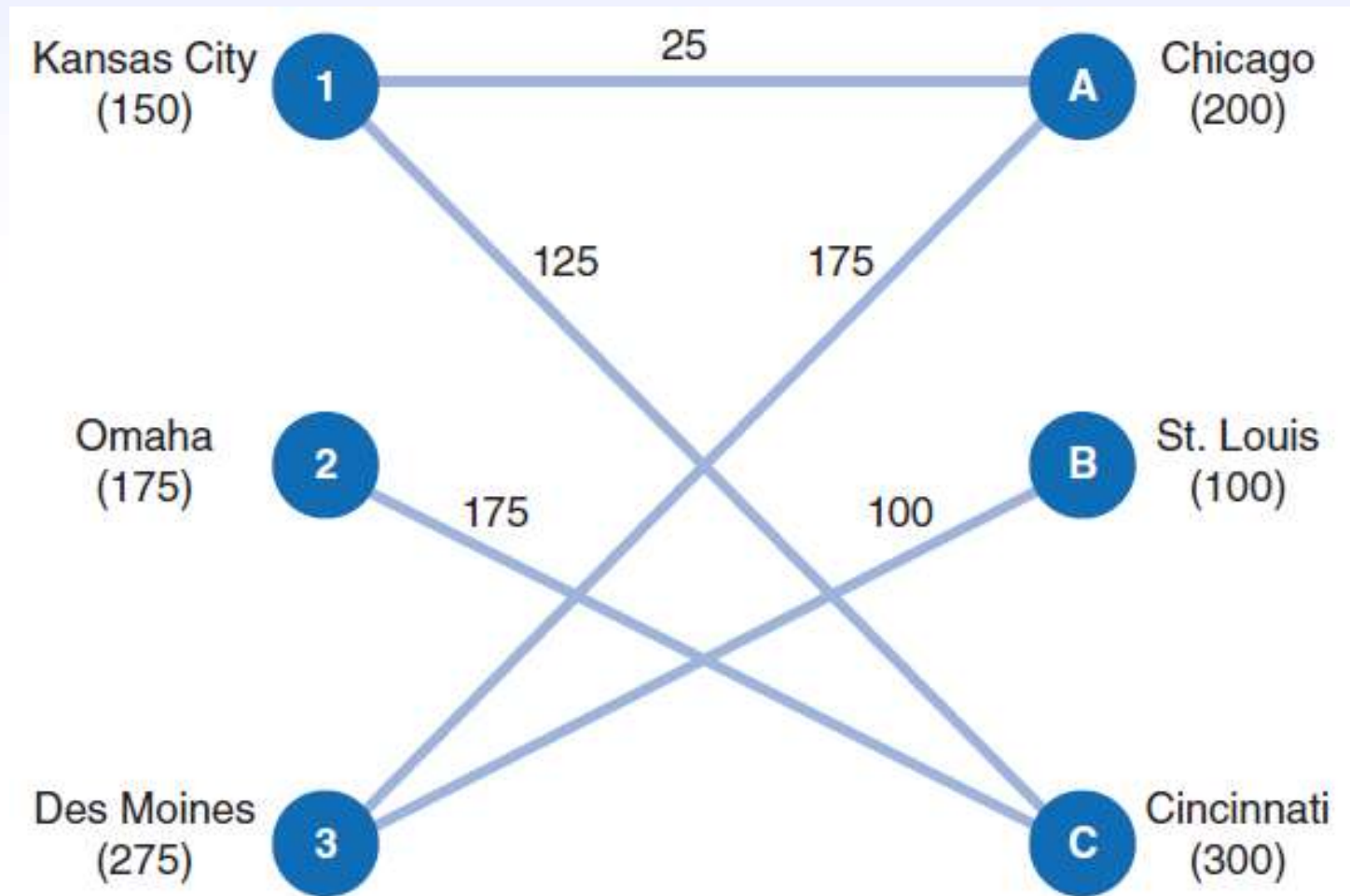
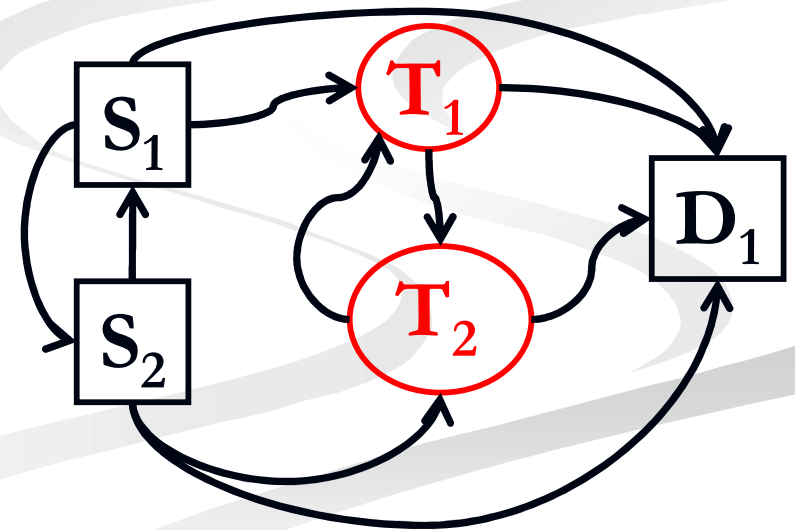


Figure 6.2 Transportation network solution for wheat-shipping example

The Transshipment Model

Characteristics

- Extension of the transportation model.
- Intermediate transshipment points are added between the sources and destinations.
- Items may be transported from:
 - Sources through transshipment points to destinations
 - One source to another
 - One transshipment point to another
 - One destination to another
 - Directly from sources to destinations
 - Some combination of these



Transshipment Model Example 2 (Solved)

Problem Definition and Data

Extension of the transportation model in which intermediate transshipment points are added between sources and destinations.

Shipping Costs

Farm	<u>Grain Elevator</u>		
	3. Kansas City	4. Omaha	5. Des Moines
1. Nebraska	\$16	10	12
2. Colorado	15	14	17

The wheat is then shipped to the mills in Chicago, St. Louis, and Cincinnati. As shown in the following Figure.

Transshipment Model Example

Transshipment Network Routes

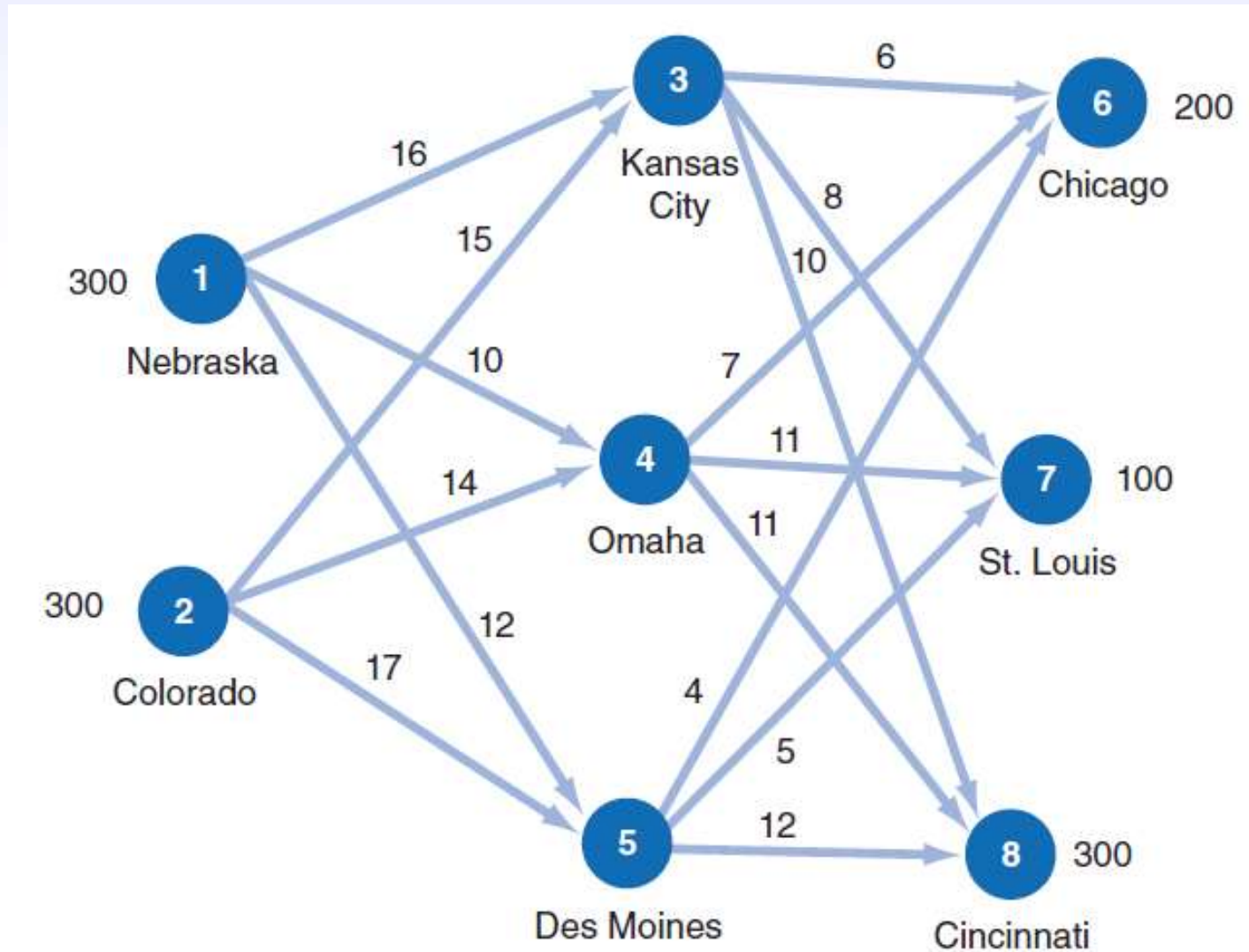


Figure 6.3 Network of transshipment routes

Transshipment Model Example

Model Formulation

$$\begin{aligned}\text{Minimize } Z = & \$16x_{13} + 10x_{14} + 12x_{15} + 15x_{23} + 14x_{24} \\ & + 17x_{25} + 6x_{36} + 8x_{37} + 10x_{38} + 7x_{46} + 11x_{47} \\ & + 11x_{48} + 4x_{56} + 5x_{57} + 12x_{58}\end{aligned}$$

subject to:

$$x_{13} + x_{14} + x_{15} = 300$$

$$x_{23} + x_{24} + x_{25} = 300$$

$$x_{36} + x_{46} + x_{56} = 200$$

$$x_{37} + x_{47} + x_{57} = 100$$

$$x_{38} + x_{48} + x_{58} = 300$$

$$x_{13} + x_{23} - x_{36} - x_{37} - x_{38} = 0$$

$$x_{14} + x_{24} - x_{46} - x_{47} - x_{48} = 0$$

$$x_{15} + x_{25} - x_{56} - x_{57} - x_{58} = 0$$

$$x_{ij} \geq 0$$

Supply constraints for farms
in Nebraska and Colorado

Demand constraints at
the Chicago, St. Louis
and Cincinnati mills

Transshipment Model Example

Computer Solution with Excel (1 of 3)

=SUM(B6:B7)

Objective function

The screenshot shows an Excel spreadsheet titled 'Exhibit6.11 - Microsoft Excel'. The formula bar displays the objective function formula: `=SUMPRODUCT(B6:D7,I6:K7)+SUMPRODUCT(C13:E15,J13:L15)`. The spreadsheet is divided into several sections:

- Farms:** A table with 2 rows (1. Nebraska, 2. Colorado) and 4 columns (3. Kansas City, 4. Omaha, 5. Des Moines, Supply). The 'Supply' column shows 300 for both farms.
- Grain Elevators:** A table with 4 rows (3. Kansas City, 4. Omaha, 5. Des Moines, Demand) and 4 columns (6. Chicago, 7. St. Louis, 8. Cincinnati, Grain Shipped). The 'Grain Shipped' column shows 300 for all three elevators.
- Mills:** A table with 4 rows (6. Chicago, 7. St. Louis, 8. Cincinnati, Demand) and 4 columns (9. Des Moines, 10. St. Louis, 11. Cincinnati, Grain Shipped). The 'Grain Shipped' column shows 300 for all three mills.
- Shipping costs:** A table with 4 rows (3. Kansas City, 4. Omaha, 5. Des Moines, Demand) and 4 columns (6. Chicago, 7. St. Louis, 8. Cincinnati, Grain Shipped). The 'Grain Shipped' column shows 300 for all three mills.
- Transshipment flows:** A table with 4 rows (3. Kansas City, 4. Omaha, 5. Des Moines, Demand) and 4 columns (6. Chicago, 7. St. Louis, 8. Cincinnati, Grain Shipped). The 'Grain Shipped' column shows 300 for all three mills.
- Cost array:** A table with 4 rows (3. Kansas City, 4. Omaha, 5. Des Moines, Demand) and 4 columns (6. Chicago, 7. St. Louis, 8. Cincinnati, Grain Shipped). The 'Grain Shipped' column shows 300 for all three mills.

Callouts point to the following formulas and data ranges:

- `=SUM(B6:B7)` points to the 'Supply' column of the Farms table.
- `=SUM(B6:D6)` points to the 'Supply' column of the Farms table.
- `=SUM(C13:C15)` points to the 'Grain Shipped' column of the Grain Elevators table.
- `=SUM(C13:E13)` points to the 'Grain Shipped' column of the Mills table.
- `Cost arrays` points to the 'Grain Shipped' column of the Mills table.
- `Constraints for transshipment flows; i.e., shipments in = shipments out` points to the 'Demand' row of the Grain Elevators table.

Exhibit 6.11

Transshipment Model Example

Computer Solution with Excel (2 of 3)

Transshipment
constraints in
cells C20:C22

The Solver Parameters dialog box is shown with the following settings:

- Set Objective:** \$C\$24
- To:** ☐ Max ☒ Min ☐ Value Of: 0
- By Changing Variable Cells:** \$B\$6:\$D\$7,\$C\$13:\$E\$15
- Subject to the Constraints:**
 - \$F\$6:\$F\$7 = \$E\$6:\$E\$7
 - \$C\$17:\$E\$17 = \$C\$16:\$E\$16
 - \$C\$20:\$C\$22 = 0
- ☒ Make Unconstrained Variables Non-Negative
- Select a Solving Method:** Simplex LP
- Solving Method:** Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Buttons: Add, Change, Delete, Reset All, Load/Save, Options, Help, Solve, Close.

Exhibit 6.12

Transshipment Model Example

Network Solution for Wheat Shipping (3 of 3)

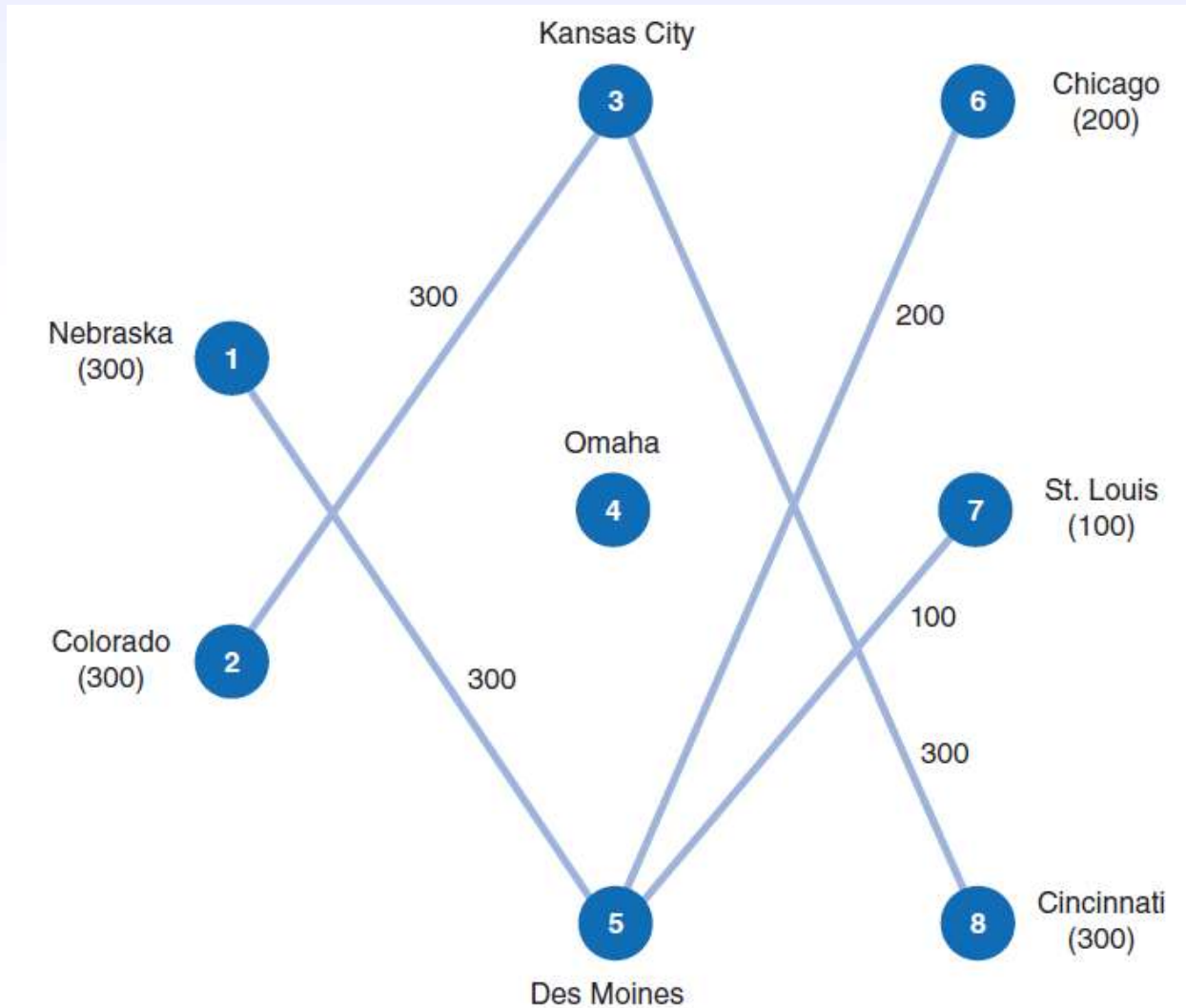


Figure 6.4 Transshipment network solution for wheat-shipping example

Example Problem Solution (3)

Transportation Problem Statement

A concrete company transports concrete from three plants to three construction sites. The supply capacities of the three plants, the demand requirements at the three sites, and the transportation costs per ton are as follows:

Plant	Construction site			Supply (tons)
	A	B	C	
1	\$ 8	\$ 5	\$ 6	120
2	15	10	12	80
3	3	9	10	80
Demand (tons)	150	70	100	

Determine the linear programming model formulation and solve using Excel.

Example Problem Solution

Model Formulation

$$\text{Minimize } Z = \$8x_{1A} + 5x_{1B} + 6x_{1C} + 15x_{2A} + 10x_{2B} + 12x_{2C} \\ + 3x_{3A} + 9x_{3B} + 10x_{3C}$$

subject to:

$$x_{1A} + x_{1B} + x_{1C} = 120$$

$$x_{2A} + x_{2B} + x_{2C} = 80$$

$$x_{3A} + x_{3B} + x_{3C} = 80$$

$$x_{1A} + x_{2A} + x_{3A} \leq 150$$

$$x_{1B} + x_{2B} + x_{3B} \leq 70$$

$$x_{1C} + x_{2C} + x_{3C} \leq 100$$

$$x_{ij} \geq 0$$

Example Problem Solution

Computer Solution with Excel

Chapter 6-Example - Microsoft

File Home Insert Page Layout Formulas Data Review View

Paste Cut Copy Format Painter Clipboard

Arial 10 A A B I U Font

Wrap Text Merge & Center Alignment

General \$ % Number

Conditional Formatting Format as Table

B10
$$=8*B5+5*C5+6*D5+15*B6+10*C6+12*D6+3*B7+9*C7+10*D7$$

	A	B	C	D	E	F	G	H	I	J
1	Example Problem									
2										
3		Construction Site								
4	Plant	A	B	C	Supply	Transported				
5	1	70	30	20	120	120				
6	2	80	0	0	80	80				
7	3	0	0	80	80	80				
8	Demand	150	70	100						
9	Transported	150	30	100						
10	Cost =	2830								
11										